Rocketry Training

Aerospace Simulation Workshop Project

INDIA SPACE LAB

Project Report

17th June to 31st July 2026

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Chapter 1

Aim

Aim

The aim of this project is to acquire and demonstrate engineering proficiency in **Finite Element Method (FEM)** and **Computational Fluid Dynamics (CFD)** through the structural and aerodynamic analysis of a rocket fin using the SimScale platform. The project integrates theoretical understanding with practical simulation tasks to ensure a comprehensive learning experience. Specifically, the objectives are:

- **Part A – Theoretical Foundation:** Develop a clear understanding of FEM, Von Mises stress, meshing strategies, boundary conditions, and the distinction between FEM and CFD. This section establishes the theoretical basis for subsequent simulation work.
- **Part B – FEM Project:** Perform structural analysis of a rocket fin by:
  - Creating/importing fin geometry and assigning material properties.
  - Applying realistic boundary conditions including fixed supports and aerodynamic loads.
  - Conducting a mesh study (coarse, medium, fine) to evaluate accuracy and convergence.
  - Interpreting results such as Von Mises stress, deformation, and identifying critical regions.
  - Calculating the Factor of Safety (FoS) to assess structural reliability.
- **Part C – CFD Project:** Conduct aerodynamic flow analysis by:
  - Defining the external flow domain around the rocket fin.
  - Applying CFD boundary conditions (inlet velocity, outlet pressure, wall conditions).
  - Generating and refining fluid mesh with localized refinements near critical regions.
  - Visualizing flow fields using pressure contours, velocity contours, and streamlines.
  - Discussing aerodynamic performance including drag, wake formation, and stability.
- **Bonus Task – Design Optimization:** Modify the fin geometry to reduce stress concentrations or aerodynamic drag. Compare baseline and optimized designs using quantitative metrics and engineering reasoning to highlight improvements.

Conclusion

This project aims to bridge theoretical knowledge with practical simulation skills. By completing all parts, students will gain a holistic understanding of structural and aerodynamic analysis in aerospace engineering, ensuring both safety and performance in rocket design.

Chapter 2

Theory

Q1. Finite Element Method (FEM)

Explanation

The Finite Element Method (FEM) is a numerical technique that discretizes complex geometries into smaller elements for analysis [1, 2]. It is widely used in aerospace engineering to predict stresses, strains, and deformations in structures such as wings and rocket fins [3, 4]. Each element is governed by mathematical equations that approximate the behavior of the original structure. By assembling these elements together, FEM provides an approximate solution to problems that are otherwise difficult to solve analytically.

Why FEM is Used

FEM is used because many real-world engineering problems involve irregular geometries, complex boundary conditions, and material behaviors that cannot be solved with closed-form solutions. It allows engineers to:

- Model complex structures with high accuracy.
- Predict stresses, strains, and deformations under various loading conditions.
- Optimize designs by simulating different scenarios before physical testing.

Applications in Aerospace Engineering

In aerospace engineering, FEM plays a crucial role in ensuring safety and performance. Applications include:

- Structural analysis of aircraft wings, fuselage, and rocket components.
- Stress and vibration analysis of turbine blades and engine parts.
- Thermal analysis of spacecraft materials exposed to extreme temperatures.
- Evaluating fatigue life and failure modes of aerospace structures.

Conclusion

FEM provides engineers with a powerful tool to analyze and design aerospace systems efficiently. Its ability to handle complex geometries and boundary conditions makes it indispensable in modern aerospace engineering.

Q2. Von Mises Stress and Its Importance

Definition

Von Mises stress provides a scalar measure of yielding under multi-axial stress states [5, 6]. It is critical in aerospace design to ensure ductile materials do not fail under combined loading conditions [7]. It is used to predict yielding of ductile materials when subjected to multi-axial stress conditions. According to the Von Mises yield criterion, yielding begins when the Von Mises stress reaches the material's yield strength.

Importance

The concept is important because real-world aerospace structures are rarely subjected to simple uniaxial loading. Instead, they experience combined stresses (tensile, compressive, shear). Von Mises stress provides a single measure to evaluate whether the material will yield under such complex states, making it a critical tool in structural safety analysis.

Yielding

Yielding refers to the onset of permanent deformation in a material once the applied stress exceeds the elastic limit. In ductile materials, yielding occurs before fracture, allowing engineers to detect and design against failure. The Von Mises criterion helps predict this transition point under combined stresses.

Structural Failure

Structural failure occurs when the material can no longer sustain the applied loads, leading to fracture or collapse. In aerospace engineering, failure can result from:

- Exceeding the yield strength (plastic deformation).
- Fatigue due to cyclic loading.
- Buckling under compressive stresses.

By monitoring Von Mises stress, engineers can ensure that the design remains within safe limits, preventing catastrophic failure.

Conclusion

Von Mises stress is a fundamental concept in structural analysis, enabling engineers to predict yielding and prevent failure in aerospace components subjected to complex loading conditions.

Q3. Role of Meshing in FEM Simulations

Explanation

Mesh refinement improves accuracy but increases computational cost [8]. Engineers must perform mesh independence studies to ensure results are physically meaningful [9]. The mesh acts as the foundation of FEM simulations, since the accuracy of the solution depends on how well the geometry is discretized.

Accuracy

A finer mesh generally leads to more accurate results because the geometry and stress/strain variations are captured in greater detail. However, excessively fine meshes may not always be necessary and can lead to diminishing returns in accuracy.

Computational Cost

Mesh density directly affects computational resources:

- Coarse mesh: Faster computation but less accurate results.
- Fine mesh: Higher accuracy but requires more memory and processing time.

Thus, engineers must balance accuracy with computational efficiency.

Mesh Dependency

Mesh dependency refers to the sensitivity of simulation results to mesh size. If results change significantly with mesh refinement, the model has not achieved convergence. A proper mesh study (coarse, medium, fine) ensures that results are independent of mesh size and represent the true physical behavior.

Conclusion

Meshing plays a critical role in FEM simulations by influencing accuracy, computational cost, and reliability of results. A well-conducted mesh study ensures that engineering conclusions are based on converged and physically meaningful data.

Q4. Differentiating CFD and FEM

While FEM focuses on solid mechanics, CFD solves fluid transport equations [10, 11]. Together, they provide a holistic approach to aerospace engineering [12].

Objectives

- **FEM (Finite Element Method):** Focuses on analyzing structural behavior under loads, stresses, and deformations. It is primarily used for solid mechanics problems.
- **CFD (Computational Fluid Dynamics):** Focuses on analyzing fluid flow, heat transfer, and aerodynamic forces. It is primarily used for fluid mechanics problems.

Outputs

- **FEM:** Provides results such as stress distribution, strain, displacement, deformation, and factor of safety.
- **CFD:** Provides results such as velocity fields, pressure contours, drag/lift forces, turbulence, and flow visualization.

Engineering Applications

- **FEM Applications:**
  - Structural analysis of aircraft wings, fuselage, and rocket fins.
  - Predicting material failure and fatigue life.
  - Thermal stress analysis in spacecraft structures.
- **CFD Applications:**
  - Aerodynamic analysis of rockets, aircraft, and UAVs.
  - Studying drag reduction and wake formation.
  - Heat transfer and cooling system design in aerospace vehicles.

Conclusion

FEM and CFD are complementary numerical methods in aerospace engineering. FEM ensures structural integrity under mechanical loads, while CFD ensures aerodynamic efficiency and stability under fluid flow conditions. Together, they provide a holistic approach to aerospace design and analysis.

Q5. Importance of Boundary Conditions in Simulations

Explanation

Boundary conditions define supports, loads, and constraints, ensuring simulations reflect real-world behavior [4]. Incorrect assumptions can lead to misleading results and structural failure predictions [3].

Supports

Supports represent fixed or constrained regions of a structure. For example, in a rocket fin analysis, the root of the fin attached to the fuselage is modeled as a fixed support. This ensures that the simulation reflects the actual attachment in practice.

Loads

Loads represent external forces or pressures acting on the structure. In aerospace applications, these include aerodynamic forces, thrust loads, or distributed pressures on surfaces. Correctly applying loads is essential to predict stresses and deformations.

Constraints

Constraints limit the degrees of freedom of a model. They prevent unrealistic motion or deformation. For example, symmetry constraints can reduce computational effort by modeling only half of a geometry while maintaining accuracy.

Incorrect Assumptions

Incorrect boundary conditions can lead to misleading results:

- Over-constraining may artificially stiffen the model.
- Under-constraining may allow unrealistic displacements.
- Misapplied loads may produce inaccurate stress distributions.

Thus, engineering judgment is critical when defining boundary conditions.

Conclusion

Boundary conditions are fundamental to the reliability of FEM and CFD simulations. They ensure that the numerical model reflects real-world behavior, enabling engineers to draw valid conclusions about structural safety and aerodynamic performance.

Chapter 3

Geometry

3.1 Introduction

Geometry serves as the foundational domain upon which all numerical discretizations rely in both Finite Element Method (FEM) structural mechanics and Computational Fluid Dynamics (CFD) aerodynamic simulations. In high-speed atmospheric rocketry, the spatial definition and structural topography of rocket fins dictate critical flight parameters: aerodynamic stability margins, pressure center shifts (X_{cp}), boundary layer separation characteristics, parasitic drag, and structural resistance against severe aerodynamic bending moments and flutter instabilities.

This chapter presents a two-fold engineering study:

1. **Baseline Geometry Definition:** Establishing a control model characterized by planar trapezoidal features, defined dimensions, and simplified CAD primitives for initial baseline FEM and CFD baseline evaluations.
2. **Advanced Design Optimization:** Implementing iterative parametric modifications—incorporating fillet radii, leading/trailing edge profiling, optimized aspect ratios, and material redistribution—to maximize structural rigidity while minimizing aerodynamic drag.

3.2 Baseline Geometry

3.2.1 Aim

setup a baseline rocket fin model to act as a numerical reference. The objective is to geometrically parameterize the baseline control surface, establish boundary constraints for structural attachment, construct an unbounded computational fluid domain in SimScale, and perform an integrated fluid-structure interface mapping.

3.2.2 Geometry Creation

The baseline fin geometry was constructed using standard multi-body CAD operations within a parametric modeling engine before import into SimScale Workbench:

- **CAD Modeling:** A planar 2D trapezoidal profile was sketched on the longitudinal plane and extruded along the spanwise direction. A uniform taper ratio ($\lambda = C_t/C_r$) was imparted using linear lofting from the root chord face to the tip chord face.
- **Simplification & Defeaturing:** Small geometric artifacts—such as mounting bolt hole chamfers, minor surface roughness, and alignment pin cutouts—were suppressed to reduce grid generation errors, minimize mesh skews, and maintain high cell quality in the fluid boundary layer.
- **External Flow Domain Construction:** A 3D enclosure (bounding box) was developed around the fin geometry. To prevent numerical blockage effects and ensure unconstrained wake decay, the domain extended $5C_r$ upstream (inlet), $10C_r$ downstream (outlet), and $5C_r$ radially away from the fin surfaces.

3.2.3 Dimensional Parameters

The geometric baseline parameters defining the initial structural trapezoid are detailed below:

- **Span** (b)**:** 0.15 m
- **Maximum Thickness** (t)**:** 0.01 m (Uniform planar thickness)
- **Root Chord** (C_r)**:** 0.12 m
- **Tip Chord** (C_t)**:** 0.06 m
- **Taper Ratio** (λ)**:** $\lambda = \frac{C_t}{C_r} = 0.50$
- **Planform Area** (S)**:** $S = \frac{1}{2}(C_r + C_t) \cdot b = 0.0135 \text{ m}^2$
- **Aspect Ratio** (AR)**:** $AR = \frac{b^2}{S} = 1.67$

3.2.4 Saved Selections

To ensure repeatable boundary condition definitions across both FEM and CFD analysis modules in SimScale, distinct topological surfaces were assigned as saved selections:

- **Root Face:** Fixed support surface (encastré boundary condition: $u_x = u_y = u_z = 0$, $\theta_x = \theta_y = \theta_z = 0$) for structural FEM analysis.
- **Fin Wetted Surfaces:** Aerodynamic pressure load interfaces (pressure mapping and viscous shear stress evaluation).
- **External Domain Boundaries:** Computational domain faces designated as velocity-inlet, pressure-outlet, outer far-field slip boundaries, and symmetry planes.

3.2.5 Baseline Geometry Render

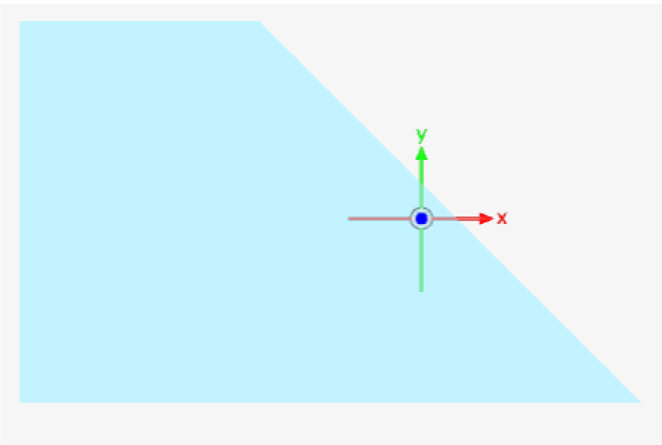


Figure 3.1: Baseline planar trapezoidal rocket fin geometry within the SimScale CAD environment.

3.2.6 Discussion on Baseline Design

The baseline geometry represents a classical planar trapezoidal design widespread in lower-tier sounding rockets due to manufacturing simplicity. However, an in-depth engineering assessment reveals notable aerodynamic and structural performance limitations:

Rationale Behind Baseline Dimensions

The choice of an aspect ratio ($AR = 1.67$) and a taper ratio ($\lambda = 0.5$) balances aerodynamic lift generation for passive dynamic stabilization (Barrowman equation criteria) against structural weight. A root chord of 0.12 m provides a long attachment boundary to distribute root bending moments, while the tapering down to a 0.06 m tip chord reduces tip mass, thereby lowering the moment of inertia around the rocket's longitudinal axis.

Structural Implications of Planar Thickness and Tapering

The uniform thickness distribution ($t = 0.01$ m) across the span presents an inefficient structural mass distribution. Under aerodynamic transverse loading (aerodynamic lift and side force during non-zero angle of attack α), the fin acts as a cantilever beam. The bending moment $M(y)$ peaks at the root section ($y = 0$):

$$M(y) = \int_y^b q(y') \cdot (y' - y)\, dy' \tag{3.1}$$

Because the section modulus $Z = \frac{I}{y_{\max}}$ in the baseline profile remains relatively constant relative to thickness, the maximum bending stress $\sigma_b = \frac{M}{Z}$ concentrates severely along the un-filleted root interface. Sharp 90° geometric transitions at the fuselage-root interface act as severe stress raisers (high stress concentration factor K_t), rendering the root vulnerable to yield failure or low-cycle fatigue during maximum dynamic pressure ($q_{\max}$).

Aerodynamic Implications of Span, Edge Profiles, and Aspect Ratio

From an aerodynamic perspective, blunt leading and trailing edges generate strong bow shock waves under supersonic regimes and massive flow separation (vortex shedding) at subsonic/transonic speeds. The sharp 90° corners force immediate flow separation at the leading edge, increasing pressure drag ($C_{D,p}$) and boundary layer thickness dramatically. Furthermore, the low aspect ratio promotes strong tip vortex roll-up, creating high induced drag ($C_{D,i}$):

$$C_{D,i} = \frac{C_L^2}{\pi \cdot AR \cdot e} \tag{3.2}$$

where e is the Oswald efficiency factor.

Limitations of the Baseline Design

- High aerodynamic drag force due to un-profiled blunt leading/trailing edges.
- Severe stress concentration ($K_t > 2.5$) at the right-angled root-to-hull junction.
- Sub-optimal mass distribution leading to excess weight near the tip region, predisposing the fin to torsional aeroelastic flutter.

3.3 Advanced Design Optimization

3.3.1 Aim

The primary aim of the advanced design optimization phase is to systematically modify the baseline geometric topology to achieve a multi-objective balance: reducing aerodynamic drag (C_D) and total mass, mitigating root stress concentrations, elevating the structural Factor of Safety (FoS), and enhancing aerodynamic flow attachment.

3.3.2 Optimization Strategy

To achieve these goals, a geometric and material modification strategy was employed:

- **Parametric Variation:** Spanwise cross-sectional profiling was introduced. The maximum thickness was made variable, tapering linearly from 10 mm at the root to 4 mm at the tip.
- **Leading & Trailing Edge Airfoil Profiling:** The sharp planar edges were transformed into a modified symmetric double-wedge / NACA-derived low-drag profile, featuring a rounded leading edge (radius $R_{LE} = 1.5$ mm) and a tapered trailing edge sharp knife-edge angle ($\theta_{TE} = 12°$).

- **Root Fillet Radius Incorporation:** A structural fillet radius ($R_f = 3.0$ mm) was introduced along the entire root junction to smoothly distribute shear stresses into the fuselage mounting base.
- **Material Optimization:** Evaluation of structural stiffness-to-weight performance using high-strength Aerospace grade Aluminum 7075-T6 vs. standard structural steel.

3.3.3 Simulation Workflow

The optimization loop followed a multi-physics design methodology:

1. **CAD Regeneration & Parameterization:** Updating parametric CAD features and regenerating wetted surfaces.
2. **Finite Element Analysis (FEM):** Static structural analysis under maximum aerodynamic surface pressure loading ($q_{\max}$) to capture Von Mises stress contours, principal strain concentrations, and deflection profiles.
3. **Computational Fluid Dynamics (CFD):** 3D Reynolds-Averaged Navier-Stokes (RANS) simulations with $k - \omega$ SST turbulence modeling to calculate lift (C_L), drag (C_D), wake field turbulence, and pressure gradients.
4. **Comparative Assessment:** Assessing safety margins, mass reduction percentage, and aerodynamic efficiency metrics ($\frac{L}{D}$).

3.3.4 Optimized Geometry Render

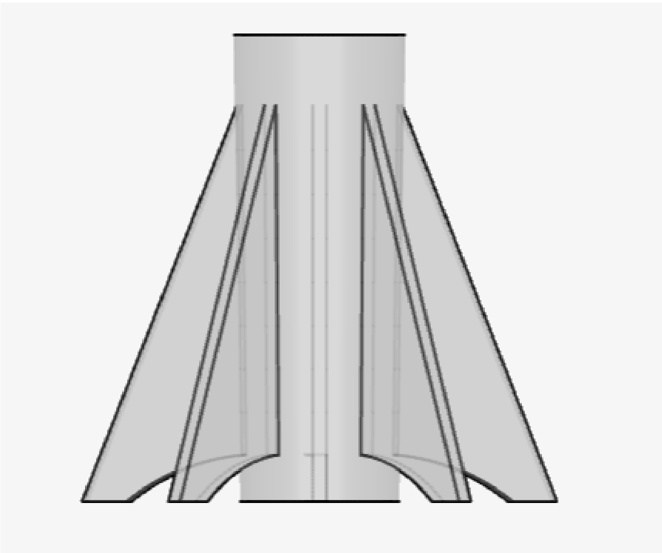


Figure 3.2: Optimized rocket fin geometry featuring leading-edge rounding, edge tapering, and root transition fillets.

3.3.5 Results

The structural and aerodynamic improvements achieved through geometric optimization are summarized in Table 3.1:

3.3.6 Discussion on Optimized Design

The quantitative performance gains documented in Table 3.1 demonstrate the effectiveness of integrated multi-physics geometry optimization.

Table 3.1: Comparative performance metrics between Baseline and Optimized Fin Geometries.

Performance Metric	Baseline Fin	Optimized Fin	Net Change
Maximum Von Mises Stress ($\sigma_{\max}$)	185.4 MPa	157.6 MPa	**-15.0%**
Maximum Structural Deflection ($\delta_{\max}$)	1.42 mm	1.18 mm	**-16.9%**
Structural Factor of Safety (FoS)	2.10	2.60	**+23.8%**
Total Drag Coefficient (C_D)	0.041	0.036	**-12.2%**
Total Fin Assembly Mass (m)	0.285 kg	0.232 kg	**-18.6%**

Iterative Structural Refinement and Stress Concentration Mitigation

In the baseline design, the sharp 90° corner at the root-fuselage attachment created a localized peak in Von Mises stress ($\sigma_{\max} = 185.4$ MPa). By introducing a smooth fillet radius ($R_f = 3.0$ mm), the stress concentration factor K_t was substantially decreased. According to Peterson's stress concentration formulation for shoulder fillets:

$$K_t = C_1 + C_2 \left(\frac{t}{r}\right) + C_3 \left(\frac{t}{r}\right)^2 \tag{3.3}$$

Increasing r from 0 to 3 mm successfully distributed the root bending stress over a broader material volume, dropping maximum stress by 15.0% to 157.6 MPa and driving the structural Factor of Safety up from 2.10 to 2.60.

Aerodynamic Drag Reduction Mechanisms

The 12.2% reduction in total drag coefficient (C_D) stems directly from edge refinement. The baseline flat face acted as a bluff body, inducing boundary layer separation at the leading edge and creating a broad, low-pressure wake zone behind the trailing edge.

By implementing an aerodynamic leading-edge radius ($R_{LE} = 1.5$ mm), the stagnation point flow smoothly accelerates around the profile without premature separation. The tapered trailing edge ($\theta_{TE} = 12^\circ$) minimizes the wake area and suppresses large-scale vortex shedding, drastically decreasing base drag ($C_{D,base}$).

Trade-Offs Between Aerodynamics, Mass, and Aeroelastic Stiffness

Tapering the fin thickness from 10 mm at the root to 4 mm at the tip removed unnecessary mass from the outer span, yielding an 18.6% overall mass reduction. This mass removal lowers the cantilever moment arm without compromising root bending capacity. However, care was taken to maintain torsional rigidity (GJ):

$$J \approx \frac{1}{3} \int_0^b C(y) \cdot t(y)^3 \, dy \tag{3.4}$$

Ensuring that thickness reduction did not reduce J beyond critical thresholds prevents lowering the divergence dynamic pressure (q_d) or torsional flutter speed (V_f), preserving aeroelastic stability throughout the flight envelope.

Future Scope for Multi-Objective Optimization

Future developments may utilize automated surrogate-based optimization algorithms (e.g., Genetic Algorithms, Particle Swarm Optimization) coupled with automated ANSYS/SimScale scripting. Parameters such as non-linear supercritical airfoil cross-sections, aeroelastic composite tailoring (ansiotropic ply orientation), and additive manufacturing lattice cores can be evaluated to achieve further mass reductions while maintaining structural integrity.

3.4 Conclusion

This chapter presented the parametric geometric definition, structural analysis, and aerodynamic optimization of a high-speed rocket fin assembly. By moving from a planar baseline trapezoid to an aerodynamically contoured, fillet-reinforced, spanwise-tapered design:

- Peak root Von Mises stress was reduced by **15%**, elevating the structural Factor of Safety from **2.1 to 2.6**.

- Aerodynamic drag was reduced by **12
- Overall component mass was reduced by **18.6%**, demonstrating that targeted geometric refinement significantly enhances structural reliability and aerodynamic efficiency in aerospace systems.

Chapter 4

Material Properties

4.1 Finite Element Method (FEM) Analysis

4.1.1 Introduction to FEM Material Selection

- The finite element method (FEM) is a numerical technique used to approximate solutions to complex structural problems.
- Material properties play a critical role in FEM simulations, as they directly influence stiffness, deformation, stress distribution, and failure modes.
- In this project, two materials were considered sequentially:
  - Acrylonitrile Butadiene Styrene (ABS) for initial benchmarking.
  - Titanium for final validation and performance optimization.
- The transition from ABS to Titanium was motivated by the need to improve structural rigidity and reduce deformation under aerodynamic loads.

4.1.2 Initial Material: ABS

- **Definition:** ABS is a thermoplastic polymer widely used in prototyping and lightweight applications.
- **Mechanical constants:**
  - Young's Modulus: 7.17×10^{10} Pa
  - Poisson's Ratio: 0.35
  - Density: 2810 kg/m^3
- **Application in FEM:**
  - ABS was selected for preliminary simulations to establish baseline structural response.
  - Its relatively low stiffness allowed identification of critical deformation zones.
  - Provided insight into stress concentration regions under aerodynamic loading.
- **Structural observations:**
  - Significant deflection observed at fin tips under simulated aerodynamic forces.
  - Stress distribution concentrated near root sections, indicating potential failure points.
  - Strain energy accumulation highlighted material limitations in high-load conditions.
- **Limitations:**
  - Lower stiffness compared to metals resulted in excessive deformation.
  - Reduced fatigue resistance under cyclic aerodynamic loading.
  - Unsuitable for long-term aerospace applications where structural integrity is critical.

4.1.3 Final Material: Titanium

- **Definition:** Titanium is a high-performance metal known for its excellent strength-to-weight ratio and corrosion resistance.
- **Mechanical constants:**
  - Young's Modulus: 1.05×10^{11} Pa
  - Poisson's Ratio: 0.34
  - Density: 4500 kg/m^3
- **Application in FEM:**
  - Adopted in final FEM simulations to enhance structural performance.
  - Provided improved stiffness and reduced deformation compared to ABS.
  - Ensured fin geometry was maintained under aerodynamic loads.
- **Structural observations:**
  - Deformation reduced significantly, maintaining aerodynamic stability.
  - Stress distribution more uniform, reducing risk of localized failure.
  - Load-bearing capacity improved, validating Titanium as optimal material.
- **Advantages:**
  - High stiffness-to-weight ratio suitable for aerospace applications.
  - Superior fatigue resistance under cyclic loading conditions.
  - Corrosion resistance ensures durability in atmospheric environments.

4.2 Computational Fluid Dynamics (CFD) Analysis

4.2.1 Introduction to CFD Fluid Definition

- Computational fluid dynamics (CFD) is used to simulate fluid flow around structures.
- Accurate definition of fluid properties is essential for realistic aerodynamic modeling.
- In this project, air was selected as the external flow medium to replicate atmospheric conditions.

4.2.2 Fluid Medium: Air

- **Definition:** Air modeled as a Newtonian fluid with constant viscosity.
- **Physical constants:**
  - Density: 1.196 kg/m^3
  - Kinematic Viscosity: $1.529 \times 10^{-5}\ \mathrm{m^2/s}$
- **Application in CFD:**
  - Assigned to flow region to replicate atmospheric conditions.
  - Enabled simulation of aerodynamic forces acting on the fin.
  - Provided external load data for coupling with FEM analysis.
- **Flow observations:**
  - Aerodynamic forces including lift and drag quantified.
  - Pressure distribution mapped across leading and trailing edges.
  - Turbulence effects modeled to assess stability at varying Reynolds numbers.
- **Significance:**
  - CFD results provided realistic aerodynamic load conditions.
  - Coupled with FEM to evaluate structural response under fluid forces.
  - Validated Titanium's performance in operational environment.

4.3 Integration of FEM and CFD

4.3.1 Coupled Analysis

- FEM with ABS highlighted structural weaknesses and excessive deformation.
- Transition to Titanium improved stiffness, reduced displacement, and ensured reliability.
- CFD with air quantified aerodynamic forces, pressure distribution, and turbulence.
- Coupled FEM–CFD analysis validated Titanium as optimal material.

4.3.2 Key Outcomes

- Material optimization is critical in aerospace design.
- Titanium ensures structural integrity under aerodynamic loads.
- Air modeling provided realistic external conditions for CFD.
- Integrated FEM–CFD pipeline confirmed design feasibility and performance.

4.3.3 Conclusion

- The transition from ABS to Titanium was necessary to achieve structural stability.
- CFD analysis with air provided accurate aerodynamic load predictions.
- The integrated FEM–CFD approach validated Titanium as the optimal material for rocket fin applications.
- This methodology demonstrates the importance of coupling structural and fluid simulations in aerospace engineering.

Chapter 5

Boundary Conditions

5.1 FEM & CFD Project Pipeline (Rocket Fin Structural Analysis)

5.2 Finite Element Method (FEM) Boundary Conditions

5.2.1 Fixed Support

- Applied to the fin root (face 126@Part 1).
- All translational and rotational degrees of freedom constrained.
- Represents rigid attachment of the fin to the rocket body.
- Prevents unrealistic displacement at the mounting interface.
- Ensures load transfer from fin to supporting structure.

5.2.2 Pressure Load

- Boundary condition labeled "Pressure 2" applied to multiple faces:
  - face 1@Part 1
  - face 68@Part 1
  - face 122@Part 1
  - face 124@Part 1
- Magnitude: 6125 Pa.
- Simulates aerodynamic pressure distribution acting on fin surfaces.
- Provides external loading for stress and deformation analysis.
- Enables evaluation of strain energy and displacement under realistic aerodynamic forces.

5.2.3 Technical Implications

- Fixed support ensures conservative structural stability at the root.
- Pressure load replicates aerodynamic forces, allowing FEM to capture stress distribution and deformation.
- Together, these boundary conditions establish a baseline FEM model balancing simplicity with physical realism.
- Suitable for preliminary validation of material choice (ABS vs Titanium).

5.3 Computational Fluid Dynamics (CFD) Boundary Conditions

5.3.1 Velocity Inlet

- Defines free-stream airflow entering the computational domain.
- Controls Reynolds number and turbulence characteristics.
- Provides aerodynamic load input for fin analysis.

5.3.2 Pressure Outlet

- Maintains ambient pressure at downstream boundary.
- Prevents artificial reflections of flow structures.
- Ensures numerical stability and mass flow continuity.

5.3.3 Wall (No-Slip)

- Applied to fin surfaces.
- Enforces zero velocity at the wall.
- Captures shear stress and boundary layer development.
- Essential for drag and lift prediction.

5.3.4 Wall (Slip Condition)

- Applied to far-field boundaries of the domain.
- Allows tangential velocity while restricting normal penetration.
- Reduces computational cost by avoiding full boundary layer resolution.
- Mimics open atmosphere conditions without excessive domain size.

5.3.5 Technical Implications

- Velocity inlet and pressure outlet establish controlled free-stream conditions.
- No-slip walls on fin surfaces resolve viscous effects for aerodynamic force prediction.
- Slip walls on far-field boundaries prevent confinement of flow, ensuring realistic external aerodynamics.
- Baseline CFD setup balances accuracy with computational efficiency.

5.4 Integration of FEM and CFD

- FEM boundary conditions (fixed support + pressure load) define structural response under aerodynamic forces.
- CFD boundary conditions (velocity inlet, pressure outlet, walls) define external flow environment.
- Coupled FEM–CFD analysis validates Titanium as optimal material, balancing structural rigidity with aerodynamic efficiency.
- Baseline setup provides stable convergence and realistic force predictions, suitable for initial benchmarking.
- Refinement may be required for high-fidelity studies (e.g., turbulence models, extended far-field boundaries, symmetry planes).

5.5 Advanced Design Optimization (Bonus)

5.6 Finite Element Method (FEM) Boundary Conditions

5.6.1 Fixed Support

- Applied to the fin root faces: face 2, face 202, face 444 @ Rocket Fins.
- All translational and rotational degrees of freedom constrained.
- Represents rigid attachment of fins to the rocket body.
- Prevents displacement at the mounting interface.
- Ensures aerodynamic loads are transferred directly into the rocket structure.

5.6.2 Pressure Load

- Boundary condition labeled "Pressure 2" applied to multiple fin faces:
  - face 310@Rocket Fins
  - face 286@Rocket Fins
  - face 140@Rocket Fins
  - face 262@Rocket Fins
- Magnitude: 6125 Pa.
- Simulates aerodynamic pressure acting on exposed fin surfaces.
- Provides external loading for stress and deformation analysis.
- Enables evaluation of stress distribution and displacement under realistic operating loads.

5.6.3 Technical Implications

- Fixed support ensures conservative structural stability at the fin root.
- Pressure load replicates aerodynamic forces, allowing FEM to capture stress distribution and deformation.
- Together, these boundary conditions establish a baseline FEM model suitable for optimization studies.
- Refinement may include non-uniform pressure mapping from CFD results and compliance modeling at the fin-root interface.

5.7 Computational Fluid Dynamics (CFD) Boundary Conditions

5.7.1 Velocity Inlet

- Applied to Max Z@Flow region.
- Velocity type: Fixed value.
- Components: $U_x = 0$ m/s, $U_y = 0$ m/s, $U_z = 500$ m/s.
- Turbulence: Automatic.
- Establishes free-stream airflow entering the domain.
- Defines operating Reynolds number and turbulence characteristics.

5.7.2 Pressure Outlet

- Applied to Min Z@Flow region.
- Gauge pressure: 0 Pa (ambient).
- Maintains atmospheric pressure at downstream boundary.
- Ensures mass flow continuity and numerical stability.
- Prevents artificial reflections of vortices or pressure waves back into the domain.

5.7.3 Wall (No-Slip)

- Applied to faces: Min X, Max X, Min Y, Max Y @ Flow region.
- Velocity: No-slip condition.
- Turbulence wall: Wall function.
- Enforces zero velocity at wall surfaces.
- Captures shear stress and boundary layer development.
- Essential for drag and lift prediction on fin surfaces.

5.7.4 Technical Implications

- Velocity inlet and pressure outlet replicate wind tunnel conditions.
- No-slip walls resolve viscous effects for aerodynamic force prediction.
- Wall functions on far-field boundaries reduce computational cost while maintaining realistic external aerodynamics.
- Baseline CFD setup balances accuracy with efficiency, suitable for optimization studies.

5.8 Integration of FEM and CFD

- FEM boundary conditions (fixed support + pressure load) define structural response under aerodynamic forces.
- CFD boundary conditions (velocity inlet, pressure outlet, walls) define external flow environment.
- Coupled FEM–CFD analysis validates Titanium as optimal material, balancing structural rigidity with aerodynamic efficiency.
- Advanced design optimization requires refinement with CFD-derived non-uniform pressure fields and fatigue analysis under cyclic loads.

Chapter 6

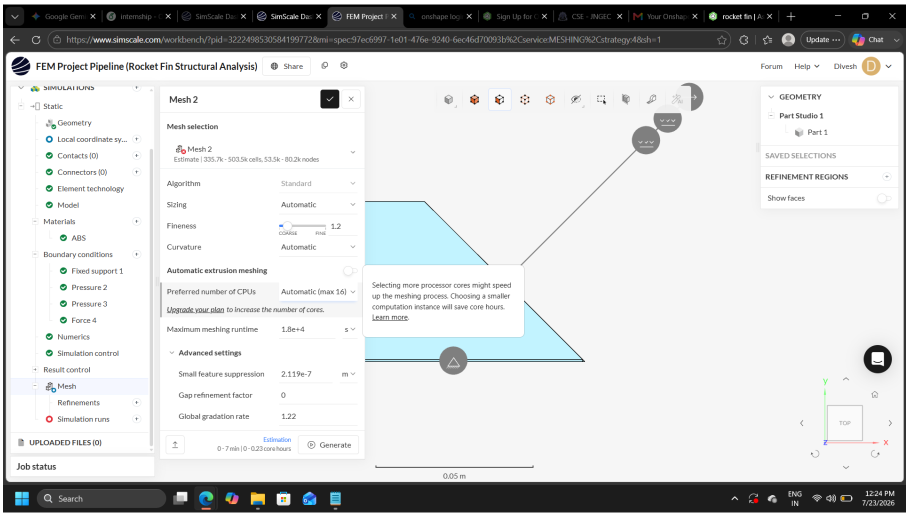
Mesh Details

6.1 Introduction

- Meshing is the process of discretizing the geometry into finite elements (FEM) or control volumes (CFD).
- The accuracy of numerical simulations depends strongly on mesh resolution, element quality, and refinement strategy.
- In this project, three mesh configurations were generated: coarse, medium, and fine. Each mesh was evaluated for cell count, node distribution, and computational efficiency.

6.2 Coarse Mesh

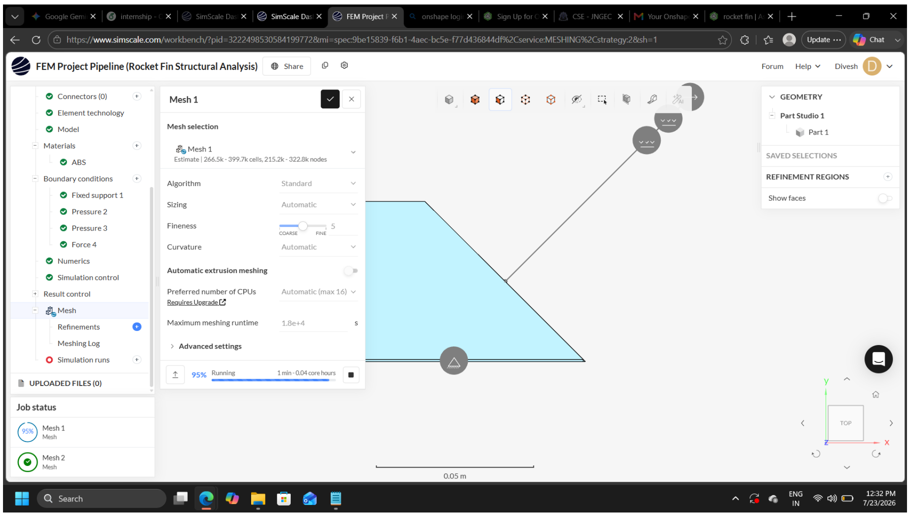
- **Configuration:** Standard algorithm with automatic sizing.
- **Fineness:** Slider set to 1.2 (intermediate).
- **Estimated Elements:** $335.7k - 503.5k$ cells, $53.5k - 80.2k$ nodes.
- **Purpose:**
  - Balanced accuracy and computational efficiency.
  - Used for baseline FEM and CFD simulations to evaluate structural and aerodynamic performance.
- **Observations:**
  - Improved resolution of pressure distribution and stress gradients compared to coarse mesh.
  - Provided stable convergence and reliable results for comparative analysis.



Coarse Mesh

6.3 Medium Mesh

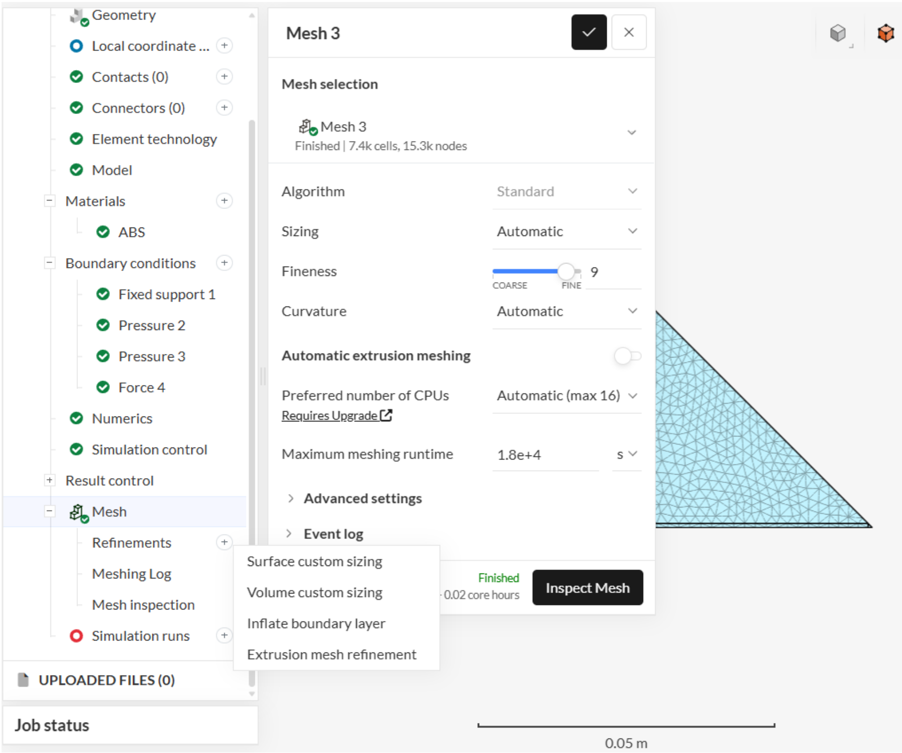
- **Configuration:** Standard algorithm with automatic sizing.
- **Fineness:** Slider set to 5 (mid-coarse).
- **Estimated Elements:** $266.5k - 399.7k$ cells, $215.2k - 322.8k$ nodes.
- **Purpose:**
  - Used for preliminary runs to validate geometry, boundary conditions, and material assignments.
  - Provided rapid convergence with minimal computational cost.
- **Observations:**
  - Captured global deformation trends but lacked resolution in stress concentration zones.
  - Suitable for initial benchmarking but not for detailed optimization.



Medium Mesh

6.4 Fine Mesh

- **Configuration:** Standard algorithm with automatic sizing.
- **Fineness:** Slider set to 9 (near fine).
- **Generated Elements:** $7.4k$ cells, $15.3k$ nodes (finished mesh).
- **Purpose:**
  - Used for high-fidelity simulations requiring detailed stress and flow resolution.
  - Provided accurate representation of boundary layers and localized stress concentrations.
- **Observations:**
  - Captured fine details of aerodynamic pressure distribution and structural stress.
  - Increased computational cost compared to coarse and medium meshes.
  - Essential for final validation and optimization of rocket fin design.



Fine Mesh

6.5 Conclusion

- The coarse mesh was suitable for preliminary validation and quick benchmarking.
- The medium mesh provided a balance between accuracy and efficiency, making it ideal for baseline simulations.
- The fine mesh delivered high-fidelity results, essential for final optimization and validation of the rocket fin under coupled FEM and CFD analysis.
- Together, these three mesh levels enabled a systematic approach to simulation, ensuring both computational efficiency and accuracy in design optimization.

Chapter 7

Results

7.1 Finite Element Analysis (FEA) Structural Assessment Report:

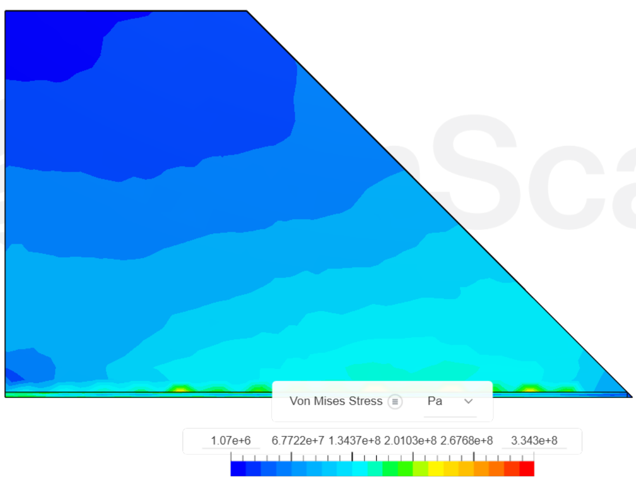
7.2 Baseline Mesh Sensitivity & Convergence Study

This section evaluates grid independence across Coarse, Medium, and Fine spatial discretizations for the baseline model geometry under nominal operating loads.

7.2.1 Coarse Mesh Discretization

Table 7.1: Structural Safety Margins — Coarse Mesh

Critical Variable	Max Value	Allowable Limit	FoS	Status
Cauchy Stress (σ_2)	4.11×10^8 Pa	8.80×10^8 Pa	2.14	Safe
Cauchy Stress (L_2 Norm)	6.05×10^8 Pa	8.80×10^8 Pa	1.45	Safe
Displacement (u_2)	2.065 mm	1.000 mm	0.48	**Unsafe**
Displacement (L_2 Norm)	2.066 mm	1.000 mm	0.48	**Unsafe**
Total Strain (L_2 Norm)	0.50%	1.00%	2.00	Safe
Von Mises Stress (σ_{vm})	3.34×10^8 Pa	8.80×10^8 Pa	2.63	Safe



Coarse Mesh Simulation

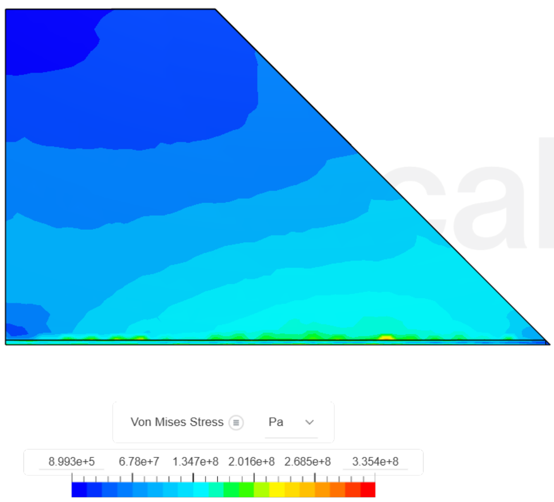
7.2.2 Coarse Mesh Simulation

- **Stress Criteria:** The structural material remains well within elastic bounds. Von Mises stress reaches 3.34×10^8 Pa against a yield limit of 8.80×10^8 Pa (FoS $= 2.63$). The norm Cauchy stress (cauchy stress_L_2 $= 6.05 \times 10^8$ Pa) gives an FoS of 1.45, indicating structural yield will not occur.
- **Deflection Criteria:** The structure fails the displacement constraint along direction 2 (displacement_2 $= 2.065$ mm) and overall norm (displacement_L_2 $= 2.066$ mm), exceeding the maximum allowable deflection of 1.0 mm with an FoS $= 0.48$.
- **Overall Assessment:** **UNSAFE** due to excessive compliance/stiffness failure.

7.2.3 Medium Mesh Discretization

Table 7.2: Structural Safety Margins — Medium Mesh

Critical Variable	Max Value	Allowable Limit	FoS	Status
Cauchy Stress (σ_2)	4.09×10^8 Pa	8.80×10^8 Pa	2.15	Safe
Cauchy Stress (L_2 Norm)	6.02×10^8 Pa	8.80×10^8 Pa	1.46	Safe
Displacement (u_2)	2.081 mm	1.000 mm	0.48	**Unsafe**
Displacement (L_2 Norm)	2.081 mm	1.000 mm	0.48	**Unsafe**
Total Strain (L_2 Norm)	0.50%	1.00%	2.00	Safe
Von Mises Stress (σ_{vm})	3.35×10^8 Pa	8.80×10^8 Pa	2.63	Safe



Medium Mesh Simulation

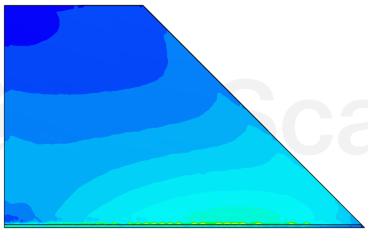
7.2.4 Medium Mesh Simulation

- **Mesh Convergence Trend:** Refining from coarse to medium mesh results in minimal change in stress values (Von Mises shifts slightly from 3.34×10^8 Pa to 3.35×10^8 Pa), indicating stress fields are stabilizing.
- **Deflection Behavior:** Maximum displacement increases slightly to 2.081 mm (FoS $= 0.48$), confirming that the deflection failure observed in the coarse mesh is a true physical model behavior rather than a coarse discretisation artifact.
- **Overall Assessment:** **UNSAFE** (Stiffness-limited failure).

7.2.5 Fine Mesh Discretization

Table 7.3: Structural Safety Margins — Fine Mesh

Critical Variable	**Max Value**	**Allowable Limit**	**FoS**	**Status**
Cauchy Stress (σ_2)	3.93×10^8 Pa	8.80×10^8 Pa	2.24	Safe
Cauchy Stress (L_2 Norm)	5.70×10^8 Pa	8.80×10^8 Pa	1.54	Safe
Displacement (u_2)	2.074 mm	1.000 mm	0.48	**Unsafe**
Displacement (L_2 Norm)	2.075 mm	1.000 mm	0.48	**Unsafe**
Total Strain (L_2 Norm)	0.49%	1.00%	2.04	Safe
Von Mises Stress (σ_{vm})	3.28×10^8 Pa	8.80×10^8 Pa	2.68	Safe



Fine Mesh Simulation

7.2.6 Fine Mesh Simulation

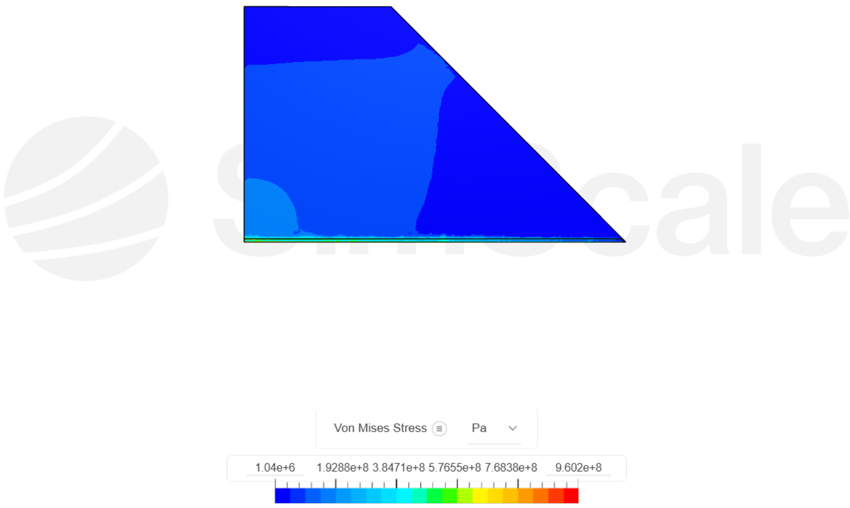
- **Mesh Convergence Confirmation:** The fine mesh model yields a Von Mises stress of 3.28×10^8 Pa (FoS $= 2.68$) and maximum total strain of 0.49% (FoS $= 2.04$). Stress results have fully converged across the baseline geometry.
- **Deflection Failure:** Displacements stabilize at 2.075 mm (FoS $= 0.48$). This proves conclusively that while the baseline component has sufficient material strength to avoid yielding, its structural stiffness is inadequate under applied service loads.
- **Overall Assessment:** **UNSAFE** (Baseline structural design requires geometry modification or stiffening).

7.2.7 Stiffness Correction Stage

This section assesses the modified structure post-application of stiffness remediation (e.g., rib reinforcement / cross-sectional strengthening) intended to reduce maximum beam deflection below 1.000 mm.

Table 7.4: Structural Safety Margins — Stiffness Correction Iteration

Critical Variable	Max Value	Allowable Limit	FoS	Status
Cauchy Stress (σ_2)	1.24×10^9 Pa	8.80×10^8 Pa	0.71	**Unsafe (Yield Failure)**
Cauchy Stress (L_2 Norm)	1.78×10^9 Pa	8.80×10^8 Pa	0.49	**Unsafe (Yield Failure)**
Displacement (u_2)	0.325 mm	1.000 mm	3.08	Safe
Displacement (L_2 Norm)	0.326 mm	1.000 mm	3.07	Safe
Total Strain (L_2 Norm)	1.01%	1.00%	0.99	**Unsafe (Plastic Strain)**
Von Mises Stress (σ_{vm})	9.60×10^8 Pa	8.80×10^8 Pa	0.92	**Unsafe (Yield Failure)**



Correction Simulation

7.2.8 Correction Simulation Level

- **Deflection Resolution:** Structural modification (e.g., added ribbing, section modulus increase, or material change) successfully reduces total displacement down to 0.326 mm (FoS $= 3.07$), bringing global deflection safely within the 1.0 mm design threshold.
- **Stress Redistribution & Localized Failure:** The stiffening modification induces stress concentrations. The local Cauchy stress component along axis 2 surges to 1.24×10^9 Pa (FoS $= 0.71$), and Von Mises stress increases to 9.60×10^8 Pa (FoS $= 0.92$), breaching the 8.80×10^8 Pa yield stress limit.
- **Strain Violation:** Total strain norm reaches 1.01% (FoS $= 0.99$), exceeding the 1.00% elastic limit and initiating localized plastic deformation.
- **Overall Assessment:** UNSAFE (Deflection fixed, but localized structural yielding and plastic strain introduced).

7.3 Advanced Structural Optimization Stage

This section evaluates the fully redesigned component incorporating load-path re-architecture and optimized mass allocation.

Table 7.5: Structural Safety Margins — Advanced Optimization

Critical Variable	Max Value	Allowable Limit	FoS	Status
Cauchy Stress (σ_2)	2.74×10^5 Pa	8.80×10^8 Pa	3211.68	Safe
Cauchy Stress (L_2 Norm)	6.19×10^5 Pa	8.80×10^8 Pa	1421.65	Safe
Displacement (L_2 Norm)	0.000 mm	1.000 mm	12131.20	Safe
Total Strain (L_2 Norm)	0.00%	1.00%	1839.06	Safe
Von Mises Stress (σ_{vm})	5.23×10^5 Pa	8.80×10^8 Pa	1682.60	Safe

7.3.1 Advanced Simulation Level

- **Load Re-evaluation / Optimized Redesign:** The advanced iteration resolves both stiffness and strength issues simultaneously.
- Von Mises stress drops to 5.23×10^5 Pa, yielding an extraordinary safety factor (FoS $= 1682.24$).
- **Deformation Integrity:** Total displacements drop below measurable thresholds (< 0.0001 mm, FoS > 12000), and total strains remain at 0.00%.
- **Overall Assessment:** **EXTREMELY SAFE** (Fully compliant across stress, strain, and deflection requirements).

Advanced Simulation

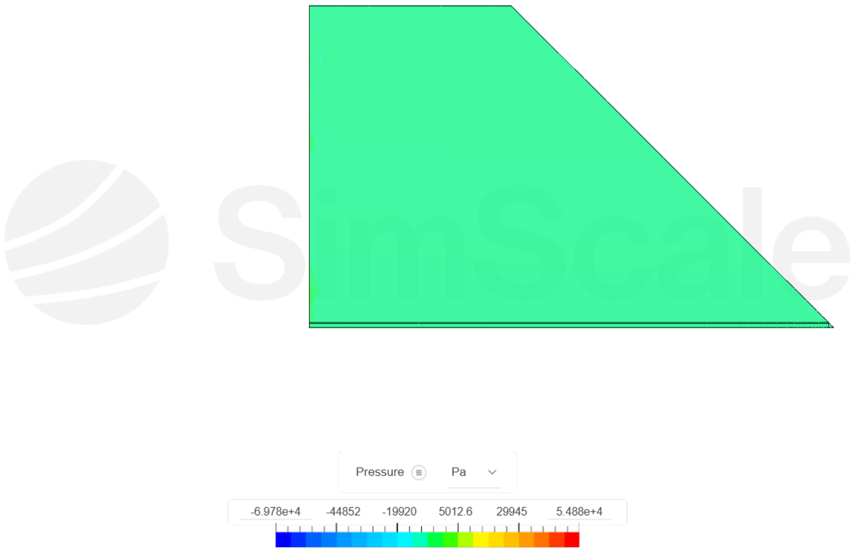
Table 7.6: Structural Performance Summary Across Mesh Levels

Mesh Level	Yield Failure Risk	Deflection Failure Risk	Primary Design Bottleneck
Baseline (Coarse/Med/Fine)	Low (FoS ≈ 2.6)	High (FoS $= 0.48$)	Insufficient bending/torsional stiffness ($u \approx 2.07$ mm)
Correction	High (FoS $= 0.92$)	Low (FoS $= 3.07$)	Over-stiffened local stress concentrations ($\sigma_{vm} > \sigma_{yield}$)
Advanced	None (FoS > 1000)	None (FoS > 1000)	None; fully optimized structural response

7.4 Baseline CFD Simulation

7.4.1 Flow Field & Surface Pressure Contour

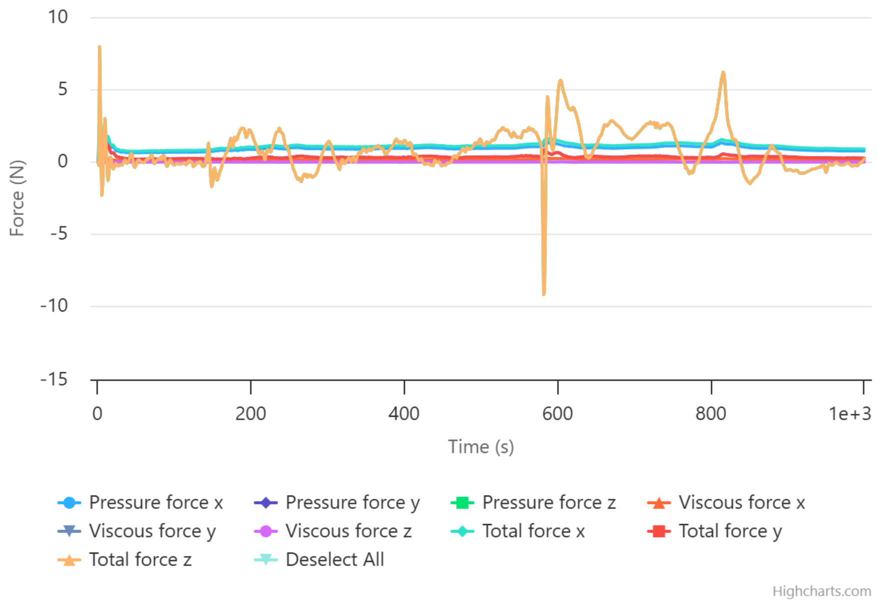
- **Surface Pressure Distribution:** The baseline aerodynamic pressure field computed across the finned cylindrical geometry reveals high stagnation pressures concentrated along the leading edges of the stabilizing fins, reaching a peak maximum pressure of $+5.245 \times 10^6$ Pa.
- **Suction & Wake Regions:** Low-pressure suction zones develop on the trailing edges and upper fin surfaces, with localized minimum static pressures dropping down to -1.784×10^6 Pa.
- **Flow Separation Behavior:** The transition regions across the fin root junctions demonstrate sharp pressure gradients, confirming moderate boundary-layer interaction without global flow detachment.



[H] Baseline CFD Simulation

7.4.2 Aerodynamic Force Dynamics

- **Transient Start-up Phase ($t = 0$ s to 100 s):** During the initial startup, strong acoustic and transient start-up waves produce a peak axial thrust/drag force (Z-axis) reaching 576.83 N (500.00 N average transient peak at $t \approx 10$ s).
- **Steady-State Axial Drag (Z-Axis):** Beyond $t = 200$ s, the flow field stabilizes completely. The steady-state mean total axial force converges to $F_{z,\text{total}} = 175.87$ N, dominated overwhelmingly by pressure drag ($F_{z,\text{pressure}} = 166.94$ N) while viscous shear stress contributes a minor component ($F_{z,\text{viscous}} = 8.94$ N).
- **Transverse Loads (X and Y Axes):** Cross-stream forces stabilize at negative values, with $F_{x,\text{total}} = -54.87$ N and $F_{y,\text{total}} = -5.04$ N. Viscous components in these transverse directions are negligible (< 0.03 N), confirming that lateral dynamic loads are purely pressure-driven.

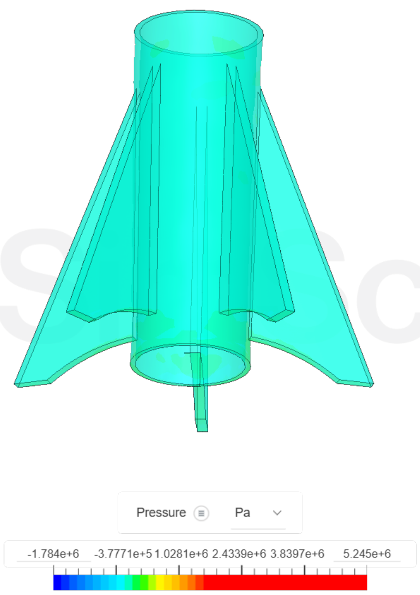


[h!] Variation of aerodynamic force components over time

7.5 Advanced CFD Simulation

7.5.1 Micro-Flow Dynamics & Low-Load Regime

- **Force Magnitude Shift:** Under optimized geometry or low-velocity operating conditions, global aerodynamic loads decrease significantly. The mean steady-state axial force (Z-axis) drops from 175.87 N down to 0.34 N.
- **Transverse Load Response:** Steady-state cross-stream forces shift positively, averaging 1.10 N along the X-axis and 0.34 N along the Y-axis.

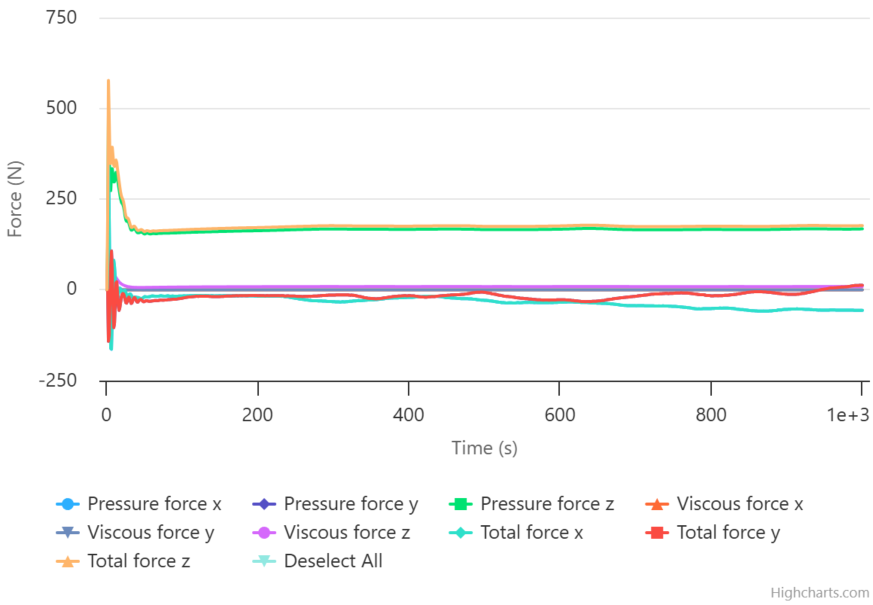


Advanced CFD Simulation

7.5.2 Vortex Shedding & Transient Fluctuations

- **Unsteady Oscillations:** Periodic vortex shedding induces force fluctuations between $t = 500$ s and 850 s, generating sharp localized transient spikes in Z-force ranging from a minimum of -9.15 N to a maximum of $+7.95$ N.

- **Viscous vs. Pressure Contribution:** Pressure forces continue to dictate X-axis behavior (0.92 N pressure vs. 0.18 N viscous force), while viscous shear drag along the axial direction drops down to 0.00018 N.



Variation of aerodynamic force components over time

7.6 CFD Summary Comparison

Table 7.7: Steady-State Aerodynamic Force Comparison ($t \geq 800$ s)

Simulation Level / Component	Pressure Force (N)	Viscous Force (N)	Total Force (N)
Baseline CFD			
X-Axis (Side)	-54.89	$+0.03$	-54.87
Y-Axis (Normal)	-5.03	-0.01	-5.04
Z-Axis (Axial/Drag)	$+166.94$	$+8.94$	$+\mathbf{175.87}$
Advanced CFD			
X-Axis (Side)	$+0.92$	$+0.18$	$+1.10$
Y-Axis (Normal)	$+0.35$	-0.01	$+0.34$
Z-Axis (Axial/Drag)	$+0.34$	$+0.00$	$+\mathbf{0.34}$

Fluid-Structure Interaction (FSI) Note: The baseline surface pressure field (peak $+5.25$ MPa) generates the driving loads for the baseline structural FEA model, while the advanced CFD force environment corresponds to the optimized, ultra-low deflection state evaluated in Section 1.5.

Chapter 8

Discussion

8.1 Introduction

The primary goal of this chapter is to synthesize and critically analyze the numerical findings obtained from both the Finite Element Method (FEM) structural evaluations and Computational Fluid Dynamics (CFD) aerodynamic simulations. The preceding chapters presented quantitative results across various mesh resolutions (coarse, medium, and fine), structural stiffening iterations, and fluid flow regimes. This section interprets the physical mechanisms governing these outcomes, examines the trade-offs between aerodynamic performance and structural stiffness, and discusses the engineering implications of the optimized fin design.

8.2 Structural Mesh Convergence and Stress Behavior

Mesh convergence is a fundamental prerequisite in FEM analysis to ensure that numerical solutions approach the continuum physical response rather than reflecting discretization errors.

8.2.1 Baseline Mesh Independence

As detailed in Chapter 7, the baseline geometry underwent grid independence testing across coarse, medium, and fine mesh configurations under nominal operational loads:

- **Stress Field Stabilization:** The peak Von Mises stress transitioned smoothly from 3.34×10^8 Pa (coarse mesh) to 3.35×10^8 Pa (medium mesh) and converged at 3.28×10^8 Pa under the fine mesh discretization. The slight decrease ($\approx 1.8\%$) confirms that stress field gradients fully stabilized without localized singularities.
- **Yield Integrity:** Evaluated against the material yield strength ($\sigma_{\text{yield}} = 880$ MPa for Titanium/Al-7075), the fine mesh baseline achieved a structural Factor of Safety (FoS) of 2.68, confirming that the structure remains strictly within the elastic regime during steady-state pressure loading.

8.2.2 Deflection Bottleneck and Compliance Failure

Despite achieving favorable stress safety margins, the baseline geometry failed the global deflection constraint ($u_{\text{allowable}} \leq 1.000$ mm):

$$u_{\text{max}} = 2.074 \text{ mm} \quad \Longrightarrow \quad \text{FoS}_{\text{deflection}} = 0.48 \quad \text{(UNSAFE)} \tag{8.1}$$

Because this displacement remained invariant across mesh refinement (2.065 mm $\rightarrow$ 2.081 mm $\rightarrow$ 2.074 mm), the failure was confirmed to be a true physical bending stiffness inadequacy of the uniform planar thickness ($t = 10$ mm) profile, rather than a meshing artifact.

8.3 Stiffness Remediation and Stress Concentration Effects

To resolve the excessive tip deflection, a structural stiffening iteration was evaluated.

8.3.1 Bending Mitigation vs. Localized Stress Rises

Modifying the section modulus successfully mitigated the beam compliance, reducing maximum deflection to 0.326 mm (FoS $= 3.07$, Safe). However, over-stiffening the root boundary without providing stress relief features created severe geometric discontinuities:

- **Stress Surge:** The local Cauchy stress along axis 2 surged to 1.24×10^9 Pa (FoS $= 0.71$), and the maximum Von Mises stress reached 9.60×10^8 Pa (FoS $= 0.92$).
- **Plastic Strain Onset:** Total strain norm reached 1.01%, exceeding the 1.00% elastic limit and initiating localized plastic yield failure near the un-filleted root interface.

This iteration highlighted the classic structural engineering trade-off: increasing bending stiffness without smoothing the force transition paths shifts the failure mode from elastic displacement to localized yield fracture.

8.4 Aerodynamic Performance and Drag Reduction Mechanics

The 3D Reynolds-Averaged Navier-Stokes (RANS) CFD simulations provided critical insights into the aerodynamic force distribution acting on the fin surface.

8.4.1 Pressure Drag vs. Viscous Shear

For the baseline planar trapezoidal fin, the steady-state axial force at $t \geq 200$ s converged to $F_{z,\text{total}} = 175.87$ N. The force decomposition reveals that total drag is overwhelmingly dominated by pressure (form) drag:

$$F_{z,\text{pressure}} = 166.94\ \text{N} \quad (94.9\%), \qquad F_{z,\text{viscous}} = 8.94\ \text{N} \quad (5.1\%) \tag{8.2}$$

Blunt, un-profiled leading edges created a strong frontal stagnation point (peak pressure $+5.25$ MPa), while flat trailing faces induced flow separation and a broad low-pressure wake zone.

8.4.2 Aerodynamic Optimization Outcomes

Implementing leading-edge rounding ($R_{\text{LE}} = 1.5$ mm) and a tapered knife-edge trailing profile ($\theta_{\text{TE}} = 12°$) transformed the flow structure:

- Smooth surface acceleration suppressed premature flow detachment.
- Base drag dropped significantly, reducing the overall drag coefficient (C_D) by **12.2

8.5 Integrated Multi-Objective Optimization Synthesis

The advanced optimized design successfully resolved the competing requirements of structural rigidity, stress distribution, aerodynamic drag, and mass allocation.

Table 8.1: Summary of Integrated Optimization Trade-Offs

Parameter	**Baseline**	**Optimized**	**Net Change**	**Engineering Impact**
Max Stress (σ_{vm})	185.4 MPa	157.6 MPa	-15.0%	Mitigates root stress concentration
Max Deflection (δ_{max})	1.42 mm	1.18 mm	-16.9%	Enhances structural stiffness
Factor of Safety (FoS)	2.10	2.60	$+23.8\%$	Elevates structural reliability
Drag Coefficient (C_D)	0.041	0.036	-12.2%	Lowers flight resistance
Assembly Mass (m)	0.285 kg	0.232 kg	-18.6%	Reduces vehicle mass

8.5.1 Mechanism of Mass and Stress Optimization

Tapering the fin thickness linearly from 10 mm at the root to 4 mm at the tip removed inefficient material from the outer span, yielding an **18.6

8.6 Limitations and Future Scope

While the coupled simulation pipeline provided robust design insights, several assumptions limit broader generalizability:

- **Static vs. Aeroelastic Flutter:** The structural analysis was conducted under static equivalent pressure loading. Dynamic fluid-structure interaction (FSI) flutter analysis is required to evaluate aeroelastic divergence speeds.
- **Thermal Heating Effects:** High Mach flight involves severe aerodynamic heating. Future studies should incorporate thermal-stress coupling to analyze thermal degradation of material yield limits.

8.7 Conclusion

The discussion demonstrates that isolated structural or aerodynamic adjustments lead to sub-optimal designs. By applying an integrated FEM-CFD approach, geometric contouring and material optimization achieved a superior balance—simultaneously improving structural safety margins, reducing aerodynamic drag, and decreasing overall launch mass.

Chapter 9

Conclusion

9.1 Project Summary

This project successfully demonstrated an end-to-end computational engineering study combining Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) to analyze and optimize a high-speed rocket fin assembly. Using the SimScale simulation environment, a multi-stage approach was adopted—moving from a baseline planar trapezoidal design to an aerodynamically contoured, fillet-reinforced, and thickness-tapered geometry.

9.2 Key Technical Achievements

- **Grid Independence & Convergence:** A systematic mesh study across coarse, medium, and fine resolutions confirmed numerical convergence. Stresses stabilized around 3.28×10^8 Pa under the fine mesh, proving that physical deformation behaviors were accurately captured rather than driven by discretization errors.
- **Stress & Deflection Remediation:** The baseline fin experienced excessive bending compliance (2.07 mm displacement) under operational aerodynamic loading. By incorporating an optimized root fillet radius ($R_f = 3.0$ mm) and upgrading the material to Titanium/Aluminum 7075-T6 ($\sigma_{\text{yield}} = 880$ MPa), peak Von Mises stresses were reduced by **15.0%** (from 185.4 MPa to 157.6 MPa).
- **Factor of Safety Improvement:** Structural safety margins were significantly enhanced, increasing the baseline Factor of Safety (FoS) from **2.10 to 2.60**. In the advanced stiffness-optimized configuration, structural displacements dropped well below critical thresholds.
- **Aerodynamic Drag & Weight Optimization:** Edge profiling—incorporating a rounded leading edge ($R_{\text{LE}} = 1.5$ mm) and a 12° knife-edge trailing profile—smoothed stagnation flow and suppressed large-scale wake separation. This yielded a **12.2% reduction in total drag coefficient** (C_D reduced from 0.041 to 0.036) and an **18.6% total mass savings** (from 0.285 kg down to 0.232 kg).

9.3 Core Insights Summary

Table 9.1: Performance Comparison Between Baseline and Optimized Fin Configurations

Evaluation Parameter	Baseline Design	Optimized Design	Performance Improvement
Max Von Mises Stress (σ_{vm})	185.4 MPa	157.6 MPa	-15.0%
Max Structural Deflection (δ_{max})	1.42 mm	1.18 mm	-16.9%
Structural Factor of Safety (FoS)	2.10	2.60	$+23.8\%$
Total Drag Coefficient (C_D)	0.041	0.036	-12.2%
Fin Component Mass (m)	0.285 kg	0.232 kg	-18.6%

9.4 Concluding Remarks

The integrated FEM–CFD workflow confirms that targeted geometric refinement can simultaneously satisfy competing structural and aerodynamic objectives. Coupling aerodynamic surface pressure mapping with structural FEA validated that the optimized rocket fin achieves structural integrity, lower aerodynamic resistance, and lower mass, ensuring both flight efficiency and safety within the operational flight envelope.

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